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“It was torture for him, and it was torture for us to see him like that”: understanding multiple ICU-related triggers of family PTSD symptoms—a qualitative thematic study

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Abstract

Purpose: Family members of patients admitted to intensive care units (ICU) are at high risk of developing post-traumatic stress disorder (PTSD) in the months following ICU discharge. We aimed to explore relatives' ICU experiences and identify common ICU-phase triggers shaping PTSD symptom experiences.

Methods: Qualitative study using in-depth, semi-structured interviews and thematic analysis. We included family members of adult ICU patients who received invasive mechanical ventilation for at least 48 h and who presented clinically significant PTSD symptoms 3 months after ICU discharge or death.

Results: Of 18 eligible family members approached, 17 agreed to participate; 8 were bereaved relatives and 9 were relatives of patients alive at 6 months. Mean age was 45 (21–70), and interviews lasted a mean of 1h38mins (49 min–2 h 35). Three main themes emerged: (1) Biographical trajectories and ICU admission, characterized by sudden illness onset, guilt, and major disruption of daily life; (2) Navigating the ICU environment, marked by distress related to the technical setting, altered relationships, vicarious suffering, and waiting; and (3) Dying and death in the ICU, describing the emotional intensity of preparation for death, interactions with medical devices, and the moment of death. PTSD symptoms appeared to arise from the accumulation of stressors throughout the illness trajectory rather than from a single event.

Conclusion: Multiple triggers related to personal history, the ICU environment, clinician communication, and end-of-life experiences were identified, highlighting the need for continuous, family-centered support throughout the ICU stay, including empathetic communication and preparation for critical transitions, to help reduce psychological distress.

Keywords: Intensive care, Family members, Qualitative research, Posttraumatic stress

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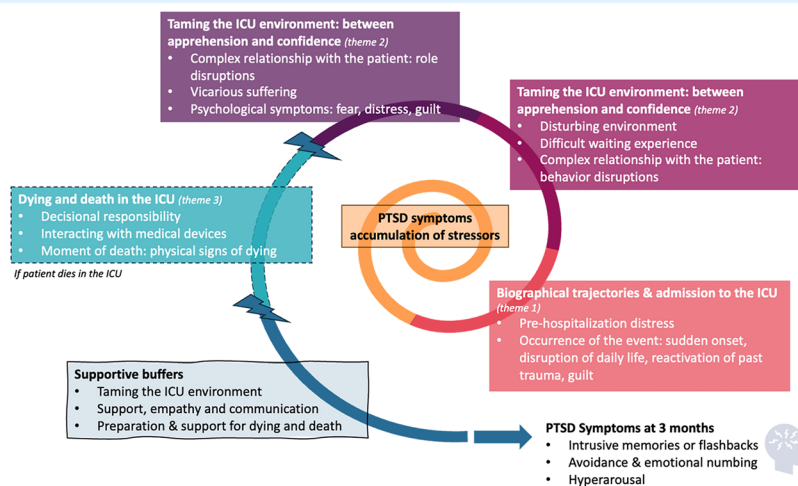
Visual abstract:

UNDERSTANDING MULTIPLE ICU-RELATED TRIGGERS OF FAMILY PTSD SYMPTOMS

A qualitative thematic study

Study design

Qualitative study using in-depth, semi-structured interviews and thematic analysis. 17 family members presenting PTSD symptoms at 3 months agreed to participate; 8 were bereaved relatives and 9 were relatives of patients alive at 6 months.



Findings

- **Family PTSD after ICU is cumulative:** driven by multiple stressors across the illness trajectory, not a single event.
- **Three major trigger domains:**
 - ICU admission & disrupted life trajectory;
 - ICU environment (technology, waiting, vicarious suffering, altered relationships);
 - Dying and death in ICU, particularly the moment of death.
- **Communication and empathy matter:** clinician support and clear information may reduce family psychological distress.
- **Implication:** Reducing family psychological distress requires continuous family-centred support that addresses communication, environmental stressors, and preparation for critical transitions and end-of-life care.

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Introduction

Having a loved-one hospitalized in the intensive care unit (ICU) can be extremely burdensome for family members,

both during the patient's stay and in the months following discharge or death [1, 2]. Numerous studies have shown that family members are at high risk of psychological

burden 1–12 months after ICU experience [3, 4]. Symptoms of anxiety and depression are frequent at 3 months [5], as are symptoms of post-traumatic stress (PTSD), reported in 10–69% of relatives [5], as well as prolonged grief when the patient dies in the ICU [6].

The cluster of psychological complications experienced by relatives of ICU patients has been described as post intensive care syndrome- family (PICS-F) [7]. Within this spectrum, PTSD represents a distinct trauma-related condition, characterized by intrusive memories or flashbacks, avoidance of reminders, persistent negative changes in mood and beliefs, emotional numbing, and heightened arousal with anxiety, irritability, and sleep disturbance. Individuals with PTSD may experience persistent distress that significantly affects daily functioning [8].

Non-modifiable risk factors include patient and family demographic characteristics, as well as clinical characteristics [9], whereas modifiable risk factors related to the ICU stay have also been identified, including communication with clinicians, emotional support, and involvement in decision-making [6, 10–12].

Because PTSD is specifically rooted in traumatic exposure and may be closely related to distressing ICU-phase experiences, qualitative research focusing on relatives with PTSD symptoms may provide valuable insights into perceived triggers and underlying mechanisms. Accordingly, this qualitative study focuses on family members who presented significant PTSD-related symptoms 3 months after ICU discharge, including some relatives undergoing bereavement. It aimed to better understand their experiences during the ICU stay and identify common ICU-phase triggers associated with subsequent PTSD symptoms.

Methods

Design

This study is part of a larger quantitative cohort study evaluating prevalence of PTSD and risk factors, in family members of ICU patients having received mechanical ventilation for a minimum of 2 days. The study was approved by a national IRB (CPP Nord Ouest I, 2022-A00902-41). Using a thematic analysis design, the aim of this qualitative component [13, 14] was to gain a deeper understanding of the experiential pathways linking ICU-related experiences to PTSD symptoms among relatives. A thematic analysis approach [15] was chosen because it is particularly suited to exploring recurring patterns in lived experiences and understanding how participants make sense of psychologically distressing events. This manuscript was prepared in accordance with the SRQR checklist [16](supplemental file 1).

Take-home message

This qualitative study of family members shows multiple PTSD triggers related to (1) biographical trajectories and ICU admission; (2) the ICU environment, including distress linked to technical setting, altered relationships, vicarious suffering and waiting; and (3) dying and death in the ICU, particularly the moment of death. PTSD symptoms result from the accumulation of stressors across the illness trajectory rather than from a single event.

Participants and data collection

Family members included in the parent cohort were enrolled after written informed consent and followed up by telephone at 3 and 6 months after ICU discharge or patient death to complete questionnaires assessing psychological symptoms. A purposive sampling approach was used to identify participants who were likely to provide rich information regarding ICU-related PTSD experiences. Specifically, at 3 months, relatives reporting PTSD-related symptoms on the PCL-5 (presence of at least one symptom endorsed in each cluster B, C, D, and E; screening for PTSD-related symptoms) were invited to participate in a telephone in-depth qualitative interview at 6 months. Among participants, some had experienced the death of their loved one in the ICU. These two groups were included as the aim of the study was to explore shared ICU-related experiences and perceived triggers of PTSD symptoms, rather than to compare outcomes according to patient survival status. Bereavement was, therefore, considered during analysis, but no formal subgroup comparisons were performed due to the qualitative design.

Interviews were conducted by a sociologist (TD) experienced in qualitative research. An interview guide was developed according to established methodological guidance [17], including literature review, preliminary guide formulation, pilot testing, and final consolidation. The guide explored participants' experiences and coping strategies across several phases—pre-ICU stress factors, ICU admission, ICU-related stress, and the post-ICU period (Table S1)—including experiences potentially relevant to PTSD symptoms. Recruitment stopped when sufficient information power was achieved, based on sample specificity, interview richness, and recurrence of relevant themes [18]. Findings related to the post-ICU phase are reported separately because of the richness of the data and the distinct processes involved.

Data processing and analysis

Interviews were audio-recorded, transcribed verbatim, and anonymized. Data analysis followed a 4-step thematic analysis process [19]. First, two authors, both sociologists, (TD and NKB) immersed themselves in the

transcripts through repeated reading, then independently generated initial codes on six transcripts. This approach enabled an in-depth examination of participants' lived experience and recurrent stressful or emotionally salient experiences associated with PTSD symptoms. Second, a preliminary codebook was developed through systematic comparison of the researchers' independent coding. Discrepancies were discussed until consensus was reached. Third, the six initial transcripts were recoded using the agreed codebook. After final consolidation, TD coded the remaining interviews. New codes could also be generated during the coding of subsequent transcripts when necessary. Finally, themes and subthemes were refined, named, and organized into overarching categories. Relevant quotes were selected to document and illustrate each theme. Although the coding process was informed by a PTSD-related conceptual focus, themes were developed inductively from the data rather than using predefined PTSD categories.

Results

Between October and December 2023, 18 family members were approached, 17 agreed to participate and one declined. Participants' characteristics are presented in Table 1. Mean age of participants was 45 (21–70) and mean duration of interviews was 1h38mins (49 min–2h35mins). Our sample included participants from 14 ICUs, 8 bereaved and 9 whose relatives were alive at 6 months. The mean length of ICU stay was 19 days.

Three themes emerged from the analysis, capturing participants' experiences during the ICU period and highlighting patterns potentially associated with PTSD symptoms (Fig. 1).

Theme 1: Biographical trajectories and admission to the ICU

Quotes for theme 1 are presented in Table 2 (Q.x). "Biographical trajectories" refers to how a person's lived experience unfolds over time.

Pre-hospitalization experiences of distress

Living with a chronic illness Participants' accounts suggest that the patient's chronic illness formed a pre-ICU backdrop of daily uncertainty and worry (Q.1,2), with progressive health deterioration further intensifying this distress (Q.3,4).

Uncertainty in the face of new symptoms Relatives describe the emergence of new symptoms in their loved one that they were unable to interpret or take seriously enough (Q.5). These symptoms created doubts and uncertainty, but not a strong awareness that something was serious -which can lead to regrets in hindsight (Q.6).

Stressful situations unrelated to the illness Other situations, unrelated to hospitalization, can also be sources of stress (Q.7), but they do not seem to affect the experience of hospitalization (Q.8). These situations are identified as burdensome by loved ones (notably work-related stress), but are put into perspective given the severity of the hospitalization: "Seeing my wife have a stroke and fall into a three-day coma put work into perspective for me" (55-0002).

The occurrence of the event: a potentially traumatic experience

A sudden onset. The event often occurs suddenly and unexpectedly (Q.9–11). This suddenness, combined with the realization of the situation's severity, can lead to negative experiences and new fears about losing a loved one: "It increases the fear of losing another relative. I started to worry about my father, my mother, my brother, my partner" (21-0007).

Disruption of daily life The ICU stay is also perceived as an upheaval in the routine of everyday life: "From the moment my father went into intensive care, my life was on standby, it was on hold" (45-0019). Organizing visits and supporting a loved one often require significant financial, professional, or geographical adjustments, impacting relatives' quality of life and causing strain and sleep problems (Q.12–14).

Past traumas Hospitalization can trigger painful memories (Q.15,16) and increase fears about the hospital or illness progression. While few relatives mentioned prior personal trauma, those who did often linked it to a difficult hospitalization or loss of a loved one. In these cases, they developed strategies to avoid the most traumatic aspects (Q.17).

Guilt and social role Hospitalization may trigger a strong sense of guilt among relatives, linked to events they feel they failed to anticipate (Q.18,19), particularly when the patient has a chronic illness or is awaiting treatment (Q.20). More broadly, some relatives feel responsible for the patient or other family members and, therefore, suppress their fears and emotions (Q.21–23), resulting in mental strain and everyday distress.

Theme 2. Taming the intensive care environment: between apprehension and confidence

Quotes for theme 2 are presented in Table 3 (Q.x).

The ICU, "Another world"

A disturbing space The technical environment of the ICU can lead to a loss of bearings, fueling feelings of insecurity and fears associated with death: "When I entered the unit, I thought, 'this is really serious, this is the beginning

Table 1 Participants' characteristics

Inclusion code	Relationship to the patient	Age	Socio-professional category	Patient status at interview	Length of ICU stay (days)	Cause of admission	PCL-5 score at 3 and 6 months
06-0014	Son	35	Employee	Alive	29	Acute respiratory distress	24 14
12-0007	Daughter	61	Executive/higher intellectual profession	Deceased in ICU	19	Multi Organ Failure	29 21
18-0015	Wife	46	Employee	Deceased in ICU	3	Acute respiratory distress	29 missing
21-0007	Sister	22	Intermediate profession	Deceased in the ICU	55	Acute respiratory distress	34 29
21-0013	Grand-daughter	22	Employee	Deceased in ICU	38	Neurological cause	33 19
22-0010	Mother	54	Employee	Alive	25	Sepsis	29 10
25-0007	Wife	55	Employee	Deceased in ICU	9	Post-op	29 23
45-0018	Son	29	Craftsperson/retailer/entrepreneur	Alive	25	Acute respiratory distress	30 24
45-0019	Daughter	21	Without a professional activity	Alive	15	Cardiorespiratory Arrest	39 35
51-0017	Husband	67	Retired	Alive	12	Shock	24 16
51-0022	Wife	56	Intermediate profession	Deceased in ICU	7	Shock	26 28
52-0017	Daughter	50	Executive/higher intellectual profession	Deceased in ICU	15	Hematologic	22 15
53-0001	Wife	70	Retired	Deceased in ICU	7	Shock	23 19
55-0002	Husband	36	Executive/higher intellectual profession	Alive	8	Sepsis	38 23
57-0007	Daughter	51	Employee	Alive	20	Hemorrhagic shock	26 22
60-0004	Daughter	43	Without a professional activity	Deceased post ICU discharge	15	Stroke	39 29
64-0004	Daughter	49	Employee	Alive	8	Stroke	46 28

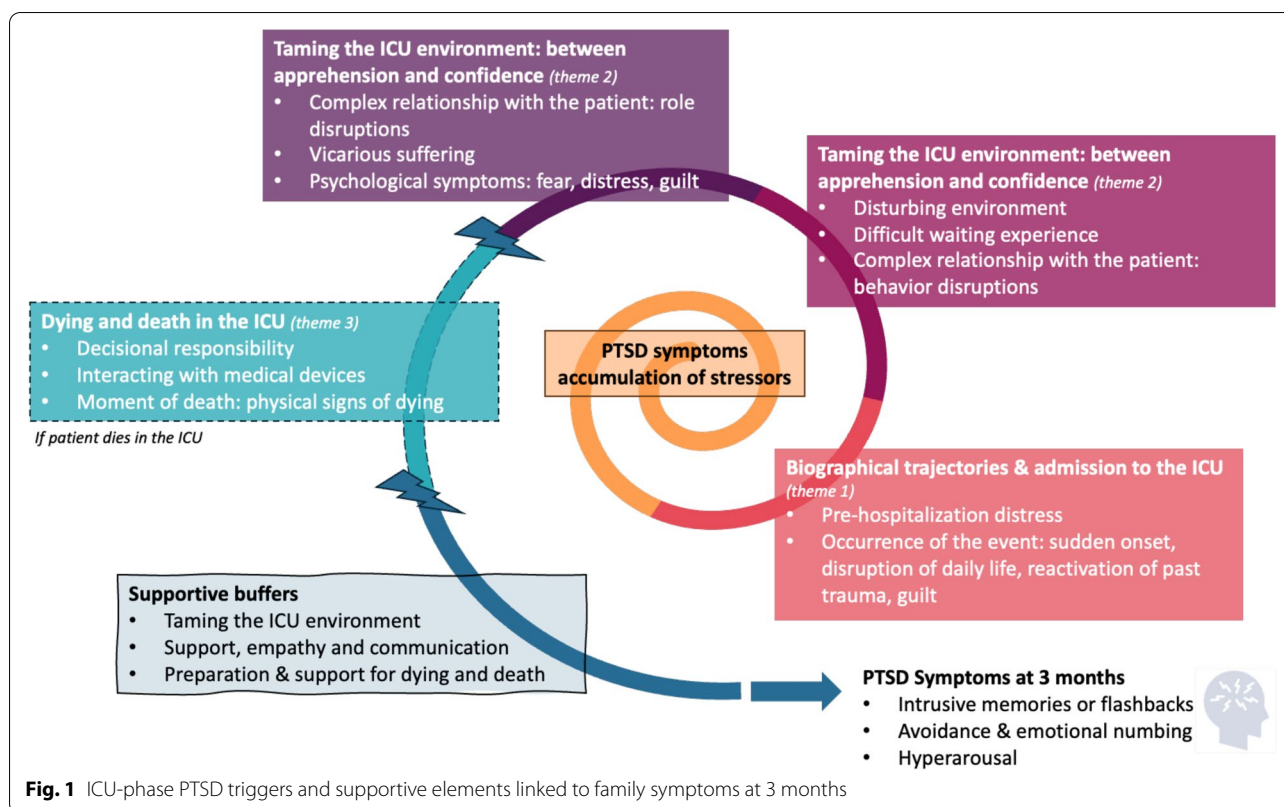
of death, it's horrible" (45-0019). Family members often use the word "violence" to describe the environment: "*Everything is violent* [in the ICU]" (25-007). The image of the patient, often sedated and intubated during the initial visits, is particularly shocking (Q.1–7). They describe seeing a dehumanized loved one and feel helpless (Q.5,7).

A complex relationship with the patient The relationship with the patient in the ICU can be complex for relatives due to physical changes and behavior disruptions (Q.7,8), feelings of loss and loneliness when the patient is unconscious (Q.9,10), and the disruption of family roles, with clinicians taking over tasks that once fell to family members, leading to feelings of rejection and loss of meaning: "I knew the nurses needed to work in peace. But still, I used to take care of my husband a lot, and I felt 'thrown away.' It's like we have no rights in

there, I don't remember what I tried to do exactly, but I was told, 'that's our job now, not yours.'" (53-0001).

Vicarious suffering Family members experience emotional suffering both personally and vicariously, witnessing their loved one's pain and distress (Q.11), as well as that of other patients, leading to significant psychological distress: "I saw some very scary things! There was a man next to us, his eyes were open, he was awake, he was intubated, and he was trying to pull out his tube, oh my, it was... So we went to talk to him with my brother because he had no family, he was all alone, and we were telling him 'you have to keep that in' and... oh no, it was awful, so upsetting!" (12-0007). This is true also in the waiting room where families witness other families' distress and sorrow (see below).

Psychological symptoms during the ICU stay Stress, lack of sleep, concerns about the patient as well as



coping with everyday life can create a high mental burden that translates into psychological and physical symptoms: “When my mom was in intensive care, I couldn’t eat; I constantly had a knot in my stomach. I was trembling all the time. It only stopped afterwards. It was the hospital; it was having to go there and see her like that that made me... I was cold, I couldn’t control my body anymore. It was constant anxiety” (12-0007).

Taming the environment

Becoming familiar with the ICU Over time, the feeling of strangeness and anxiety can diminish (Q.6, 12–14). As families become accustomed to the ICU, they tend to accept the environment and the patient’s condition (Q.12–14). The patient’s room plays a key role in separating their intimate space from the rest of the ICU, and personalizing it helps improve the ICU experience: “We redecorated the room... all the photos were great, they gave us strength” (45-0018).

Comforting spaces The ICU can also become comforting as relatives become more familiar and engaged: “It’s strange to say this, but we didn’t want him to leave intensive care. We thought if he could stay in intensive care until the end, and the day he leaves it’s to go home, we would have preferred that” (60-0014). Familiarity with

the environment and the team, as well as the proximity to technical resources, helps reassure relatives: “We knew that if anything happened, she could be taken care of quickly; that reassured us too” (22-0010).

Support in the ICU

A beneficial presence Relatives emphasized the availability of the ICU team (Q.15–19), the value of clear explanations that helped them become familiar with the unit (Q.15,16), and attentive listening as a key resource for coping (Q.17,18). The ability to contact staff at any time further contributed to feelings of support (Q.18,19).

Communication and information Families emphasized the need for clear, consistent information, as a lack of updates can cause confusion and hinder effective planning (Q.20,21).

Empathy, a pivotal behavior. Communication focused only on health issues is experienced as difficult (Q.22,23). Lack of empathy in delivering information or discussing the patient can be hard for families, leading to doubts about care quality (Q.24–26). A compassionate approach, both verbal and non-verbal, can ease the impact of difficult news: “A kind word, a little smile, a hand on the shoulder, it’s not much, but it helped me enormously” (22-0010).

Table 2 Representative quotes for Theme 1—biographical trajectories and admission to the ICU

Quote n°	Quotes	Inclusion code
<i>Pre-hospitalization experiences of distress</i>		
1	The last three months before her transplant were just pure depression... I really thought about nothing else all the time. I'd get up, get a call, and think it was about her. I really thought about nothing else	45-0018
2	He had heart problems and all that, he had a lot of health issues in recent years, but he always managed to get through it. In fact, for 4 years, he only had health problems. Every time, it was the hospital, and we kept wondering, 'How is he going to get through this?' We were always worried	60-0004
3	Over the past two or three years, he started showing symptoms of anxiety and panic. I think it was related to his illness. I believe he was scared of what was going to happen. He talked to me about it, saying he was afraid of what was coming, afraid of dying, afraid of making decisions	21-0007
4	In the last six months, she was very, very, very tired. She could barely move from her room to the kitchen, and it was hard for her even to get up to answer the door. That's when I started to worry... actually, it became an obsession	45-0018
5	My father, in the months before he ended up in intensive care, you could see he was a bit sick. He could see it himself because he was losing weight, so we were worried, but not too worried because he had these small check-ups from time to time. He would go for an X-ray, a scan, an appointment with this or that specialist. It had almost become a routine for several months, but I could see he was losing a lot of weight. He was weaker than before, but we mostly attributed it to age because he was getting past 70	06-0014
6	He had a behavior that (silence) sometimes, because he was already a grumpy person by nature... But I found him even grumpier in the last weeks, and I didn't think much about it at the time. Well, now of course I have lots of questions, but then I tell myself, 'you don't have to blame yourself'	25-0007
7	I had some stress in our relationship because my wife was working a lot, and it was hard... she didn't really listen	51-0017
8	I was already going through a stressful time when the incident with my wife happened, so it didn't exactly make things better... but as a result, I decided to change things up a bit. I changed jobs and went to work at another company	55-0002
<i>The occurrence of the event: a potentially traumatic experience</i>		
9	It happened all of a sudden... I rushed over to the hospital to be with him in the intensive care unit, and when I saw him, he was in really bad shape. He was just so weak, and suddenly he had all these machines hooked up to him	06-0014
10	On Saturday night, out of the blue, he says, 'Hey, I've got a little bump on my neck' and starts checking it out. We went to see the family doctor, and she said, 'You need to go to the ER right away.' And actually, he never came out of the hospital	51-0002
11	It actually started a week before. She had an ingrown toenail that kept getting infected. She'd seen a podiatrist who put a wick in it, but it got infected, the infection spread, and it triggered an erysipelas with a streptococcus. She ended up with a sudden and severe sepsis...	57-0007
12	During the months of hospitalization, my routine was completely turned upside down. Besides all the mental stress, there was also the physical exhaustion. Going to the hospital, sleeping there, and visiting my dad took up so much of my time	06-0014
13	What was tough at that point was not just the intensive care, but also the fact that since I lived far away, I had to figure out how to visit her every day. Financially, it was a concern too, so I ended up getting a place nearby to be close to her	45-0018
14	I already had sleep issues before, not necessarily related to this, but it was always a bit tough to fall asleep or if I woke up during the night. Now, though, I'm feeling really exhausted...	55-0002
15	I've always had an issue with hospitals... When I was a kid, I had meningitis. I don't remember it, but I was in the hospital for a long time. I don't have any clear memories of it, but I've always had a problem with hospitals, probably because of that	52-0017
16	I had a significant moment in my life in 2018 when my little sister told me that her partner had just had a heart attack. And now, with my husband's hospitalization, it's really bringing all those feelings back. Yes, it's definitely resurfacing a lot. It's also affecting my little sister, bringing her back to that time	51-0022
17	I asked my mom and brother to help support me by the arms, and I kept my eyes closed from the moment we reached the ICU doors until I was in my dad's room...	52-0017

Table 2 (continued)

Quote n°	Quotes	Inclusion code
18	We saw some things in the Hematology department that were concerning. We were a bit naive and didn't react, and maybe if we had, we could have saved some time. Perhaps we could have sorted things out in a few days, I'm not sure	21-0007
19	For the chest compressions, I wondered, 'Did I do everything right, or...?' But that was a long time after	51-0017
20	I felt selfish because I thought it was great that she had the surgery, but I didn't realize how serious the operation was. Seeing her suffer like that was truly awful; the guilt was overwhelming	45-0018
21	There were things I sensed and didn't want to show my family, which felt normal. So I was very worried... There was the fear of death... I started thinking about it as soon as the doctor told me it was very, very serious	22-010
22	I didn't really have time to fully process it... I mean, I understood what was happening, but I didn't have time to panic. I was scared, but I had to stay composed because I needed to be the adult	21-0013
23	I tried to stay strong so my family wouldn't be more worried than they already were	22-010

A difficult waiting experience

Waiting as a source of stress Waiting—due to illness progression and frequent visit delays (tests, care, transfers)—can disrupt relatives' daily lives and visit planning (Q.27–29). Delays also create uncertainty about the illness and contribute to helplessness and fatalism (Q.30). The waiting room, often isolated from the unit, can affect relatives' experience by fostering emotional contagion (Q.31,32).

The rollercoaster of hope The instability and changes in the patient's condition fuel phases of hope, uncertainty, and fatalism, which can be a mental burden (Q.33,34). These changes can also disrupt the daily lives of relatives and cause difficulties in organizing activities outside the hospital (Q.35).

Awakening phase and new stress This phase is both awaited by families (Q.27–30) and a source of concern about the patient's short- and medium-term consciousness (Q.36–37). Post-sedation delirium is particularly striking (Q.38,39), often causing disorientation, helplessness, and worries about the future.

Theme 3. Dying and death in the ICU: the unavoidable intensity beyond preparation

We interviewed eight bereaved relatives whose loved ones died in the ICU. This theme, therefore, reflects end-of-life ICU experiences in this subgroup only and not those of relatives of ICU survivors. Quotes for theme 3 are presented in Table 4 (Q.x).

Decision to withhold or withdraw life sustaining treatment (WWLST)

Decisional responsibility Families may want to convey the patient's wishes, but do not wish to decide on WWLST (Q.1,2). Clarifying that this is a medical responsibility

can ease their burden (Q.1). Some confuse withdrawal of treatment with euthanasia, highlighting communication gaps (Q.3).

The possibility of anticipation and preparation Knowing the patient will die lets families prepare emotionally and practically (Q.4). A key concern is not leaving the patient alone (Q.5). Time to arrive and be with the patient is highly valued (Q.6,7).

Dying and death

A time to die While families do not bear the responsibility for the WWLST decision, many of them are responsible for choosing the time of death: "He had told us that they had made the decision to stop the machines and that it was then our responsibility to give the "go-ahead"" (51-0002). Although difficult, allowing loved ones to choose when to stop treatment can be important, as it helps them plan visits and provide meaningful support to the dying (Q.8,9).

Interacting with medical devices Some families feel the ICU is not a suitable place for dying (Q.10). The technical environment and device management can cause ambiguity or distress (Q.11,12), and reducing treatment may disrupt the dying process's temporality (Q.13).

The moment of death Recognizing signs of death can be difficult, leaving relatives upset for not noticing (Q.14) or distressed by the signs themselves (Q.11–15). Seeing the dead body is intense, and families need more preparation and support (Q.16,17).

Discussion

Current literature on PICS-F shows that one-third of ICU patients' families are at risk of developing severe symptoms, including anxiety, depression, PTSD, or complicated grief [5]. Our study further elucidates the social and

Table 3 Representative quotes for Theme 2—taming the intensive care environment: between apprehension and confidence

Quote n°	Quotes	Inclusion code
<i>The ICU, "Another world"</i>		
1	All the devices, like I told you, the devices. It was reaching some kind of breaking point (stammers). I couldn't take it anymore, I lost who knows how many kilos. I... I just couldn't handle it...	12-0007
2	The first night, I didn't go see him because my sister was there and she told me, 'I don't recommend you go see him.' I looked at her and said, 'Why?' She said, 'Well, listen, he's hooked up to a bunch of tubes, and there's blood everywhere. I don't recommend you see him.' So, I didn't go that night, but I went the next day. The next day, it was better—he still had the tubes, but no more blood. Still, seeing him like that really shocked me	60-0004
3	It was traumatic... I mean, seeing your mother almost naked, with bruises everywhere, and her hands all dried out—it's obvious how hard that was	57-0007
4	It was torture, it was torture for him, and it was torture for us to see him like that—unable to move, unable to speak, his mouth wide open (silence), his hands tied down, without understanding what had happened, without knowing how he ended up there, or if he was going to be okay. We talked to him, and he understood us, he could hear us because he moved his eyes, so we knew... but (silence) it was incredibly hard to see him in that situation	60-0014
5	It was shocking for me to see her like that in a coma, asleep, and I didn't know if she could hear us or not. It looked like she was having really terrible nightmares—we could see it. She was crying, making faces, and it was awful because we felt so helpless	45-0018
6	I think the worst part was the intubation, seeing him with his mouth wide open and his eyes closed... After a few days, I got 'used to' seeing him like that, so it became less shocking, but it was still really hard to go through	45-0019
7	Yes, and then she had edema, she was swelling up, I didn't recognize her hands anymore — she was always so petite. I could see that it was... that her body was changing	12-0007
8	And after a while, they woke him up, but only for two days. And he couldn't talk, I'm not sure he even understood what was going on; his eyes were just... empty	21-0013
9	What's really overwhelming in the ICU is all the machines and not being able to communicate with the other person... I mean, the interaction, because I was communicating, but it was just me, all by myself	53-0001
10	I was telling her how hard it was to find myself alone in front of my mother's body	12-0007
11	He was conscious (sobbing), he was conscious, he looked at me, he couldn't really speak, he seemed extremely weakened, well (silence)... it was very upsetting	06-0014
<i>Taming the environment</i>		
12	The first time, we didn't even know what an induced coma meant, we didn't know what intubation was, we had no idea if he would ever wake up or what state he'd be in if he did. Seeing all that the first time was overwhelming, but the second time, we were a bit more prepared	06-0014
13	Unfortunately, you start to 'get used to it' It's sad, but you begin to get used to seeing all these sick people. I was shocked by all the tubes and everything, but mainly because she was between life and death	45-0018
14	At first, I couldn't even bring myself to step into the unit; I was so scared. But by the end, it's kind of strange, but I was feeling a bit better, if you can say that. Anyway, after the second day, I didn't have a choice—if I wanted to see my dad, I had to go in and visit him (sobs). And then, well, there were all the machines and everything, and I knew they were necessary	52-0017
<i>Support in the ICU</i>		
15	<i>They took the time we needed, even if it meant spending a bit more time on us, just to make sure we understood everything. So that was really helpful</i>	21-0013
16	<i>As for the teams, we have nothing but praise. They were excellent. They were very present and took the time to support us. They were truly great. They kept telling us we could call anytime, and they were always there for us</i>	52-0017
17	<i>Honestly, we had a fantastic reception... For our daughter's hospitalization, we have nothing but praise for the team. They really took great care of her</i>	22-0010
18	They were really nice, and everything went very well. They were always attentive; the nurses were great, and the doctor took the time to explain things to me. Every day when I went in with my sisters-in-law, he would update us on how she was doing and how the night had gone	55-0002
19	There was always someone to reassure me, saying, 'Yes, Mr. S., don't worry, I'll call the doctor and he'll come to see you.' The doctors were really great; they explained that they were doing new tests each time... They were very honest because whenever they didn't know something, they told me	45-0018

Table 3 (continued)

Quote n°	Quotes	Inclusion code
20	We got some info, but actually, the next day or a couple of days later, it wasn't quite the same. We were talking among ourselves, saying, 'What did you understand?' and 'What did you get?' Then sometimes we'd figure out things that others hadn't understood, or the other way around	21-0013
21	They showed us the scan, but not the whole thing, just the latest one. It wasn't clear, the information about it wasn't clear. But with the number of doctors we had in front of us, no matter what we said, if they felt it was over, it was hard to say otherwise	21-0007
22	Medically, health-wise, everything was fine, but on a personal level, it was... well, not just with the doctors, but... I don't know, it just felt a lot colder	21-0013
23	The first meeting was like a hammer blow. It was really, really harsh because the doctor said, 'Well, maybe the brain isn't too damaged, we don't know yet, but even if she wakes up, she'll need a heart transplant.' So yeah, it was clear, but it really hit me hard	51-0017
24	So she woke up, and he said to me, 'She's very, very hard to put to sleep; she resists instead of letting go.' And I felt like saying, 'Who would let go with a tube down their throat, with all these machines and all this noise?' Who would let go?	12-0007
25	Well, I got all the information, but it was all so overwhelming, so violent	25-0007
26	It was extremely difficult to meet with the doctors. One day, I asked if I could see the doctor the next day, and they told me, 'Yes, yes, in the early afternoon.' I arrived very early because the hospital parking lot is packed from 1:30 p.m. onwards. I got there at 1 p.m., and by 7 p.m., the doctor still hadn't come to see me. I left at 7 p.m. without having seen him. I spent six hours in the waiting room... all for nothing, I mean he could have shown up just briefly, just to show me that I matter	53-0001
<i>A Difficult waiting experience</i>		
27	We were living day by day. We didn't know what was going to happen, and we didn't want to plan ahead because it was too uncertain. So yeah... I don't know. Everything was hard	52-0017
28	It was very, very—(stumbles)—really difficult, super stressful, it was a situation where... we didn't know what was going on anymore. We didn't understand anything, and anyway, there was nothing else we could do	51-0017
29	Anyway, what more could we expect? (crying) We were just there, waiting. The care wasn't for us	52-0017
30	I wanted to know everything right away, but no one really had any answers because he had only been on the machines for a few hours. We didn't really know how he would react in the short or long term. Of course, all the questions I had—whether he would survive, whether he would recover, etc.—at that moment, neither the nurses nor the doctors could give me an answer	45-0019
31	In the waiting rooms of the ICU, we saw other people who were in the same situation as us—waiting, struggling, suffering. There's something very intense that happens between families in those moments... and it really affected one of my daughters	22-0010
32	What's really painful is in the waiting room... At that moment, there are a lot of tears. Many people are crying constantly; you can see they're not doing well, which is understandable. There's so much sadness, people with their heads down, there's no joy at all... It's a place full of sorrow. It's a place with a lot of crying, and sometimes you hear screams. That really traumatized me; it was very upsetting	45-0018
33	For two weeks, it was like that—shaking, a knot in my stomach, wondering, 'What state will I find mom in?' and there was no improvement, sometimes it was worse, sometimes a little better, so there was a glimmer of hope, and then in the evening, it would all fall apart again	12-0007
34	We didn't know if she was conscious or if her brain was functioning properly; we didn't know any of that. We were in uncertainty for a long time, for several days. That was really tough	51-0017
35	There were quite a few setbacks and alerts where I had to hit the road again because I'm 500 km away and it's a 5-h drive. I had to make emergency trips back and forth, driving at night, in the evening, etc., because the emergency services had transferred her from the nursing home to the hospital or from rehabilitation back to regular hospitalization. It was like a rollercoaster, and it was exhausting	60-0004
36	What worried me was the waking up because I kept thinking, 'How is she going to be when she wakes up with that tube?' Knowing my mother, I thought it was impossible for her to accept that. I wanted her to wake up, but at the same time, seeing her condition, I wondered, 'What state will she be in afterward?' If it's just to end up in a vegetative state...	12-0007
37	We didn't know if she was conscious or if her brain was working properly. We didn't know any of that. We were in uncertainty for quite a while, for several days. That was really tough. It was difficult, and we tried to talk to her, to see... to bring up memories without pushing her too much	51-0017
38	And once, we were called because she absolutely wanted to go to the bathroom and she was screaming, so the doctor came. He told her 'no,' and she said, 'If you don't take me to the bathroom, I'll tell my daughters that you tried to rape me.' That's where we had gotten to!	57-0007
39	Waking up was difficult because, actually, I was with her eldest son, and he was insulting us. He didn't know where he was and ended up insulting us	51-0022

Table 4 Representative quotes for Theme 3—dying and death in the ICU

Quote n°	Quotes	Inclusion code
<i>Decision to Withhold or Withdraw Life Sustaining Treatment (WWLST)</i>		
1	They told me, "We need to make the decision to stop the machines." So I told the doctor, actually we weren't married, we were just in a civil union, so I wanted the opinion of his children. And the doctor said, "No, but anyway, it's not the family that makes the decision, it's us." Well, I said, "That way, at least, no one has to make a decision, no one will blame anyone, no one will hold anything against anyone." I thought, "That's perfect."	51-0022
2	The doctor told me, 'When we see from her results that her illness is taking over, we'll stop the treatments and let her go gently, sedated so she doesn't suffer.' And that's what happened. I'm glad they made that decision	12-0007
3	And they called my grandmother to tell her that he was having too much trouble breathing and that, if we agreed, they were going to euthanize him, well, not really euthanize him but help him pass away. (...) I believe they overdosed him on medication	21-0013
4	They still gave me time to come to terms with it (silence) without forcing it on me abruptly	25-0007
5	I arrived first, actually I ran there because I thought it was really urgent and I didn't want him to die alone, it's the one thing I didn't want	21-0013
6	We asked the nurses around 8:30 PM, and she said, 'I don't think he'll make it through the night.' So we called everyone who had left—children, grandchildren—and got everyone back together. We all gathered around my father, and he passed away with his children, grandchildren, and wife by his side	21-0013
7	They let me stay with him overnight. The nurses came in with flashlights to avoid turning on the lights and waking me up. Honestly, it was really kind of them. I spent a beautiful, loving night with my dear, and it wasn't horrible. It was my last night, knowing he was going to die, but we truly shared a lovely night together (silence). So, I want to keep that memory of love and connection with him	51-0002
<i>Dying and death</i>		
8	If I had told him 'no, I'm not ready yet, give me half an hour more,' he would have accepted it... He gave me the instruction to do it in the evening, between such and such a time, let's say, before midnight. He didn't impose 10 PM and say 'alright, let's disconnect and that's it.' Not at all. He let me come to him when it was the right moment	25-0007
9	He said "When you're ready, you call us." And it was when we were all ready that the nurse came to sedate him and stop the infusion. So, we made the decision of when and we were with him	53-0001
10	It shocked me because..., well, but I don't know... I mean, I can't judge care in this situation, but to me, well there was nothing left in the room except the bed, it felt like very bare	21-0007
11	I didn't handle it very well, actually, because... even though the doctor had explained it to me, he had explained it. He had turned off the monitors before we arrived. And there was just one screen in the antechamber so he could see when the heart stopped beating. It was so we wouldn't hear the beep, but I didn't handle it well because there was a final gasp, and I jumped. Honestly, I was scared. I was on the verge of fainting, actually. It was my sister who... who held me up. It was... yes, it was very overwhelming	51-0002
12	They hadn't turned off the monitor; they had left it visible. So I was looking at the monitor, and the nurse could see that I was watching it and completely falling apart, and I kept saying, 'This isn't possible, this isn't possible.' My father told the nurse to remove the monitor. The nurse asked him, 'Why?' and didn't do it right away. She said, 'Why?' and stood her ground. (...) So she turned the monitor so I couldn't see it, and after that, it still took several minutes before his heart stopped	21-0007
13	So, I thought it would be over in a few minutes, but it was longer than I expected, and it was very, very stressful because on one hand, we didn't want him to go, but on the other hand, we were also eager for it to be over for him. And actually, there were moments when I think they had lowered the oxygen, or I don't know, we felt like he wasn't breathing anymore, and then suddenly he would start again, and then it would stop	21-0013
14	It was in the middle of the night. I didn't realize that Mom had passed away, and then two nurses came into the room and asked us to step outside. Afterward, they told us, 'Your mother has passed.' And I said, 'Oh, really? When?' I thought they were still treating her, but actually, she had been gone for ten minutes, and I didn't know... I hadn't realized	12-0007
15	But there were moments when I thought it was over, and actually, it wasn't at all. And even after he had passed, apparently there were spasms, or muscles twitching, I don't know, and it felt very strange	21-0013
16	Afterward, you see a person who is lifeless, so it's not the same... and the care that's given isn't the care done at the funeral home. This was really... yes, it was shocking. And I had never seen that before, it was the first time	21-0007
17	For her, it was obvious that families wanted to see the body at that moment (silence), I found myself all alone in the room. I barely dared to approach, but I did, I touched him, but it shocked me. I had never touched a dead person in my life. I thought, 'Wow, he's so cold, he's so stiff,' but it was horrible (silence). It shocked me (silence), and the smell, all of that, those are things that stay with me	25-0007

NB: Theme 3 pertains specifically to end-of-life ICU experiences within the bereaved subgroup

relational pathways emerging during the ICU phase that help contextualize PTSD symptoms among relatives at 3 months, using a qualitative and narrative approach that complements categorical and cross-sectional measures.

ICU-based PTSD triggers identified in this study arise not only from the ICU environment and illness severity, but also from disruptions in daily life, role changes, waiting-related stress [20], and vicarious suffering. These reflect a sequence of ICU experiences [21–23]—including adapting to the technical environment, witnessing patient changes, and interacting with the medical team—that may contribute to PTSD symptoms. While some participants were bereaved, the ICU-phase distress described in the narratives may overlap with grief-related processes, particularly after death, and should, therefore, be understood as part of a broader post-ICU psychological burden that includes PTSD experiences. Our findings also underscore the ambivalent role of the ICU, which may both intensify psychological burden and provide support. However, as the present analysis included only relatives with significant PTSD symptoms at 3 months, these supportive aspects should be interpreted as coexisting with, rather than preventing, PTSD-related distress.

Triggers for PTSD symptoms were observed at every stage of the illness trajectory. The first is often the sudden and unexpected occurrence of the event [21, 24], even in chronic illness contexts. The onset of illness triggers a biographical transition [25, 26], disrupting daily life and future plans, generating fear, distress, and guilt [27, 28], reconfiguring roles, and reducing occupational and leisure activities [29]. Early support from family and healthcare professionals is, therefore, crucial to limit disorientation and emotional distress.

Although pre-existing stressors were reported, they often receded into the background during ICU admission. Thus, PTSD symptoms may be understood as emerging from the interplay between the ICU experience and prior vulnerabilities, with biographical factors contributing to differential susceptibility rather than acting as primary drivers.

Following the initial shock, ICU admission introduces additional stressors related to illness severity, prognostic uncertainty, proximity to death, and the technical environment. Clear information, staff availability, and family-centered support can help mitigate distress [30, 31], and increasing familiarity with the ICU may foster trust and a sense of safety.

Physical and behavioral changes in the patient, particularly during sedation or unconsciousness, often leave relatives feeling helpless and uncertain about how to interact [32]. Although awakening is anticipated as relief, agitation and delirium may instead be experienced as traumatic [33, 34], highlighting the need for improved

information, support [35], and appropriate family involvement [36].

While most ICU patients survive, some die, often following a decision to WWLST. Bereaved relatives reported high distress, reflecting the intertwining of PTSD and grief, particularly when unprepared or unable to say good-bye [6]. Family conferences, non-abandonment, and facilitated closure may alleviate these symptoms [37, 38]. Although families generally do not wish to bear responsibility for end-of-life decisions [39, 40], they value anticipation, time to gather and bedside presence. Adequate preparation for the dying process, including medical devices and physical signs of death, appears essential to reduce the traumatic impact.

This study contributes to a better understanding of ICU-phase triggers associated with PTSD symptoms among ICU family members up to ICU discharge or death. Further research should explore post-ICU trajectories and experiences of relatives without PTSD to identify protective factors and supportive practices.

This study has several limitations. Transferability beyond the French healthcare context may be limited. The qualitative design and voluntary participation may introduce selection bias, although refusal rates were low. Restriction to mechanically ventilated patients and French-speaking relatives may further limit representativeness. The analysis was conducted by two researchers, which may limit interpretive triangulation despite ongoing consensus discussions. Although survivor interviews touched on transfer and post-ICU experiences, these were not analyzed in the present study, which focuses specifically on ICU stay-related experiences; post-ICU triggers are addressed in a separate analysis. Finally, as the sample included only relatives with PTSD symptoms, findings reflect this subgroup and may not generalize to relatives without PTSD, whose experiences could not be examined here.

Conclusion

This study identifies multiple ICU-related triggers associated with PTSD symptoms among family members. These findings highlight the need for targeted, multidisciplinary interventions during the ICU stay, from admission through end-of-life care or transfer out of the ICU, with particular emphasis on clinician communication and psychological support. Although causation cannot be established, these associations warrant further quantitative investigation.

Supplementary Information

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Author contributions

NKB, EA, JR, FR designed the study. TD collected the data. NKB and TD analysed the data. NKB wrote the first draft of the manuscript. TD, EA, FQ, AR, VS, BE and JR revised and amended the manuscript. All authors accept responsibility for submitting the final manuscript for publication.

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Data availability

De-identified qualitative data may be made available from the corresponding author upon reasonable request and following approval by the scientific committee of the FAME project, in accordance with data protection regulations.

Declarations

Conflicts of interest

NKB, TD, AR, JR, VS and FP have no conflict of interest to declare. EA reports fees for lectures (money to him or to his research group) from Alexion, Fisher&Payckle, Mundipharma, Gilead and Pfizer.

Data reporting

The reporting of this study was guided by the Standards for Reporting Qualitative Research (SRQR) checklist (Table S2).

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